

0 8 . 1

A student investigated a method for estimating the concentration of protein in solution by using a measure of the density of the solutions.

Copper sulfate solutions of different concentration have known densities, so they can be used to measure the density of other solutions.

The student prepared a dilution series of a copper sulfate solution.

Complete **Table 3** by giving all headings, units and volumes required to make 30 cm³ of the concentration of the copper sulfate solution shown.

[2 marks]

Table 3

Concentration of copper sulfate solution / g kg ⁻¹	Volume of 100 g kg ⁻¹ copper sulfate solution / _____	Volume of water / _____
75	_____	_____

Question 8 continues on the next page

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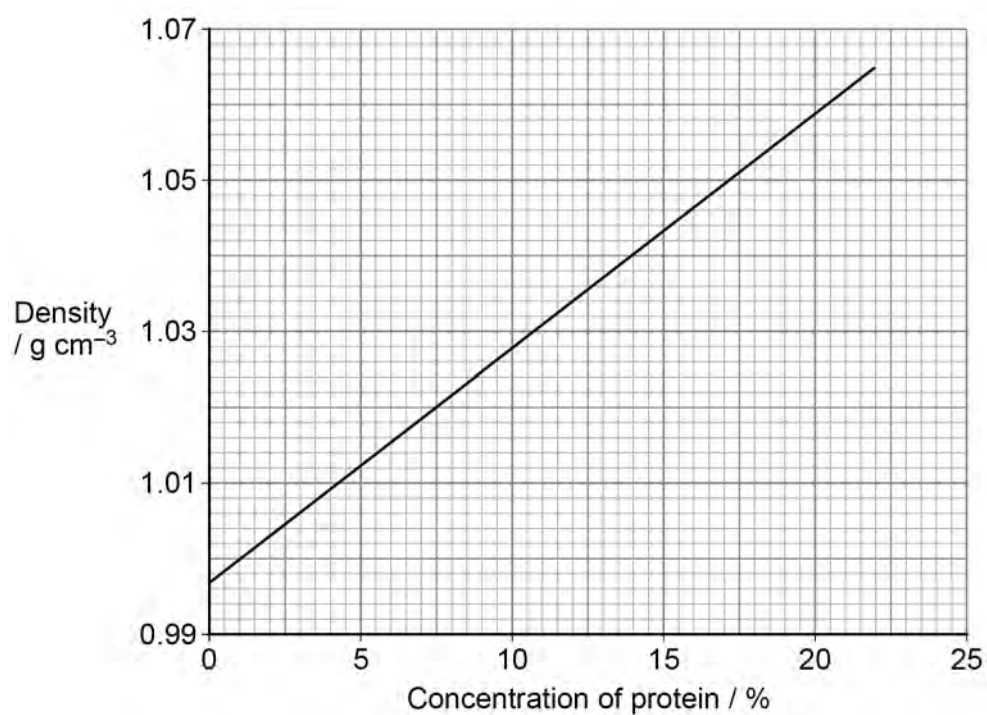
Table 4 shows the densities of the dilution series of the copper sulfate solution.

Table 4

Concentration of copper sulfate solution / g kg^{-1}	Density of solution / g cm^{-3}
0	0.997
25	1.014
50	1.030
75	1.048
100	1.065

Figure 5 shows the densities of protein solutions of different concentration.

Figure 5



Do not write
outside the
box



The student put one drop of 10% protein solution into each of the copper sulfate solutions shown in **Table 4**.

Using **Figure 5**, he predicted that the drop would sink in the 0 and 25 g kg⁻¹ copper sulfate solutions and float in the 50, 75 and 100 g kg⁻¹ copper sulfate solutions.

Give the density of the 10% protein solution **and** explain why the student predicted that the drop would sink in the 25 g kg⁻¹ copper sulfate solution.

[2 marks]

Density of 10% protein solution _____ g cm⁻³

Explanation _____

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State the range of possible concentrations of a protein solution that sinks in 75 g kg⁻¹ copper sulfate solution and floats in 100 g kg⁻¹ copper sulfate solution.

[1 mark]

Minimum concentration _____ %

Maximum concentration _____ %

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0 8 . 4

Blood donation involves healthy donors giving blood that can be used to treat hospital patients.

When donors arrive, the haemoglobin concentration of their blood is tested.

A sample of each donor's blood is added to a copper sulfate solution to determine whether the haemoglobin concentration is high enough to donate.

Errors sometimes occur with this test.

Tom has a concentration of haemoglobin high enough to donate.

Lucy has a concentration of haemoglobin too low to donate.

Evaluate the consequences of errors occurring when Tom's and Lucy's blood samples are tested.

[3 marks]

Consequences of measurement error for Tom's blood _____

Consequences of measurement error for Lucy's blood _____

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